Software Requirements Specification for:

Student attendance tracking system

Version V5

Prepared by Hlib Bokatov

Lodz university of technology

20.04.2026

Table of Contents

Contents

1. Introduction 3
   1.1. Purpose 3
   1.2. Document Conventions 3
   1.3. Intended Audience and Reading Suggestions 3
   1.4. Product Scope 3
2. Overall Description 3
   2.1. Product Perspective 3
   2.2. Product Features 4
   2.3. User Classes and Characteristics 4
   2.4. Operating Environment 4
   2.5. Design and Implementation Constraints 4
   2.6. Assumptions and Dependencies 4
   2.7. UML Class Diagram 5
3. System Features 6
   3.1. Use Case Diagram 6
   3.2. User Authentication 6
   3.3. Attendance Marking 6
   3.4. Attendance Overview 6
   3.5. Report generation 6
4. External Interface Requirements 7
   4.1. User Interfaces Overview 7
   4.2. Hardware Interfaces 7
   4.3. Software Interfaces 7

5. System Features/Modules 7
5.1.1 Description and Priority 7
5.1.2 Stimulus/Response Sequences 7
5.1.3. Activity Diagram 8
5.1.4. State Diagram 8
5.1.5. Functional Requirements 8
6. Nonfunctional Requirements 9
6.1. Performance 9
6.2. Security 9
7. GitHub repository 9

Revision History

Name	Date	Reason for Changes	Version
Hlib Bokatov	03.03.2026	Initial version	V1
Hlib Bokatov	15.03.2026	Added Use case Diagram	V2
Hlib Bokatov	24.03.2026	Added UML Class Diagram	V3
Hlib Bokatov	13.04.2026	Added Activity and State Diagram	V4
Hlib Bokatov	20.04.2026	Finished the SRS document	V5

1. Introduction

1.1. Purpose

This document specifies the software requirements for student attendance tracking system. This software is designed for automation attendance tracking for university classes

1.2. Document Conventions

Requirements are labelled as:

- FR-xx – for functional requirements
- NFR-xx – for non-functional requirements

All mandatory requirements use the word “shall”

1.3. Intended Audience and Reading Suggestions

This document is intended for:

- Developers – to implement features
- Testers – to verify system correctness
- University administration – to validate requirements

Recommended reading order:

- Introduction
- Overall Description
- System Features
- External Interface Requirements
- System Features/Models
- Non-functional Requirements

1.4. Product Scope

The Students Attendance Tracking System is a web-based application that allows:

- Teachers to mark attendance digitally
- Students to monitor their attendance percentage
- Automatic eligibility calculation for exams
- Generation of attendance reports

The system replaces manual paper-based attendance registers and improves accuracy and accessibility

2. Overall Description

2.1. Product Perspective

Students Attendance Tracking System is a new web application that shall integrate with the university’s existing student information system. It shall use a secured database to store the attendance records

2.2. Product Features

The system shall provide:

- User authentication (Student, teacher, admin)
- Attendance marked by teachers
- Attendance percentage calculator
- Eligibility determination for exams
- Attendance report export (PDF)

2.3. User Classes and Characteristics

The main user classes of the system are:

- Students - the largest user group. They log in to check their attendance, view their percentage per course, and see whether they are eligible for the final exam.
- Teachers - use the system regularly during classes to mark student attendance, edit records when needed, and export reports. They need a simple and fast interface because attendance is usually marked at the beginning of the class.
- Administrators - manage user accounts, courses, and system configuration. This is the smallest group but has the highest privilege level.

Students and teachers are the most important user classes to satisfy, since they interact with the system daily

2.4. Operating Environment

The system shall operate in the following environment:

- Server side: Linux-based server, running a Node.js runtime and a PostgreSQL database.
- Client side: any modern web browser (Chrome, Firefox, Edge, Safari) on desktop or mobile devices.

2.5. Design and Implementation Constraints

- All communication between the client and server shall be encrypted via HTTPS.
- The system shall be implemented as a web application (no native mobile app in this version).
- PostgreSQL shall be used as the database engine.
- Source code shall follow clean code principles and be maintained in a Git repository.

2.6. Assumptions and Dependencies

- It is assumed that the university provides a stable internet connection and a hosting environment for the system.
- It is assumed that each student and teacher has a valid university account.
- Any change in the university's authentication system may require adjustments on the side of this application.

2.7. UML Class Diagram

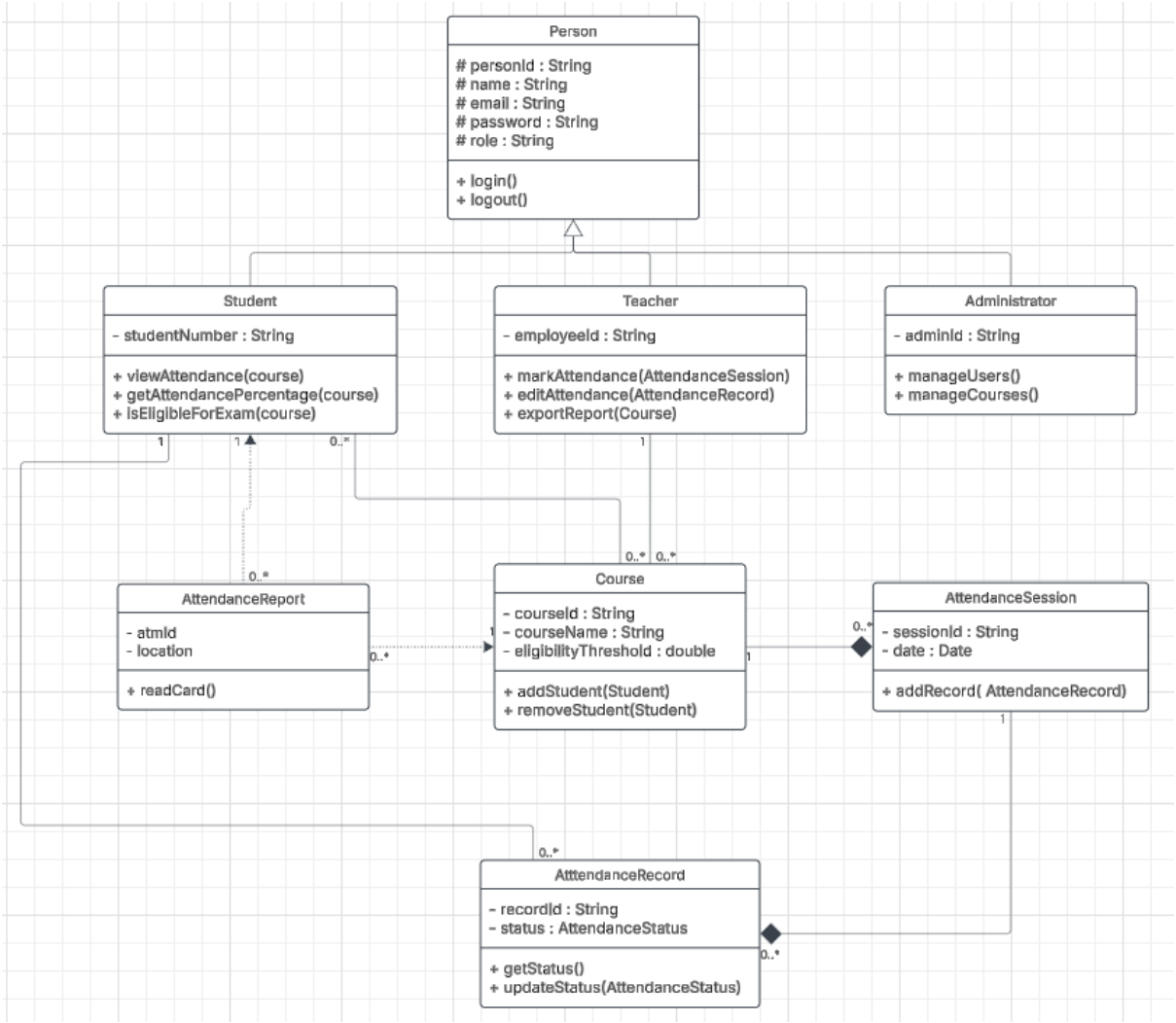

Figure 1 UML Class Diagram - Shows the main classes of the system (User, Student, Teacher, Course, Attendance, Report) and their relationships, attributes, and methods

3. System Features

3.1. Use Case Diagram

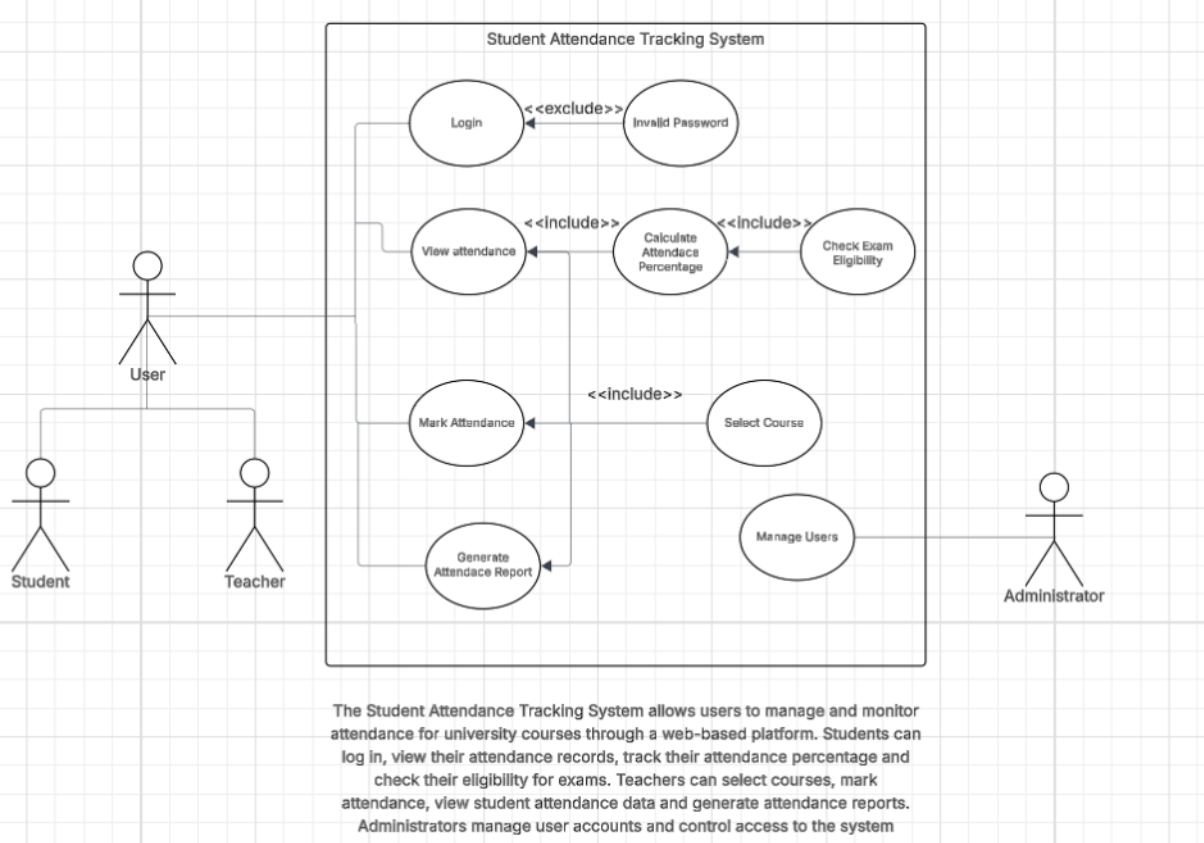

Figure 2 Use Case Diagram - Illustrates the main interactions between actors (Student, Teacher, Administrator) and the system's core functionalities such as login, marking attendance, and generating reports

3.2. User Authentication

Allows students, teachers, and administrators to log in with their university credentials and access features according to their role.

3.3. Attendance Marking

Allows teachers to mark students as present, absent, or justified during classes and save the results to the database.

3.4. Attendance Overview

Allows students to view their attendance percentage per course and check if they meet the eligibility threshold for the exam.

3.5. Report generation

Allows teachers and administrators to export attendance reports in PDF format for a selected course.

4. External Interface Requirements

4.1. User Interfaces Overview

The system shall provide a graphical web-based interface accessible from any modern browser. Users interact with the system through buttons, forms, drop-down lists and tables.

After opening the application, the user is shown a login screen with fields for username and password. Once logged in, the user is redirected to a dashboard that depends on their role:

- Students see a list of their courses and their current attendance percentage. Clicking a course opens a detailed view with each class date and status.
- Teachers see a list of the courses they teach. Selecting a course shows the list of enrolled students with buttons to mark each one as present, absent, or justified. A "save" button stores the changes.
- Administrators see a management panel to add or remove users and courses.

4.2. Hardware Interfaces

The system does not require any dedicated hardware. It runs on any device with a web browser and an internet connection, such as a laptop, desktop computer, tablet, or smartphone.

4.3. Software Interfaces

Allows teachers and administrators to export attendance reports in PDF format for a selected course.

5. System Features/Modules

5.1.1 Description and Priority

The system allows users to log in using their university credentials and restricts access based on their role. Priority: High

5.1.2 Stimulus/Response Sequences

Stimulus: user enters valid credentials and clicks "log in".

Response: system authenticates the user and opens the dashboard corresponding to the user's role.

Stimulus: user enters invalid credentials.

Response: system shows an error message and stays on the login screen.

Stimulus: user clicks "log out".

Response: system ends the session and redirects the user to the login screen.

5.1.3. Activity Diagram

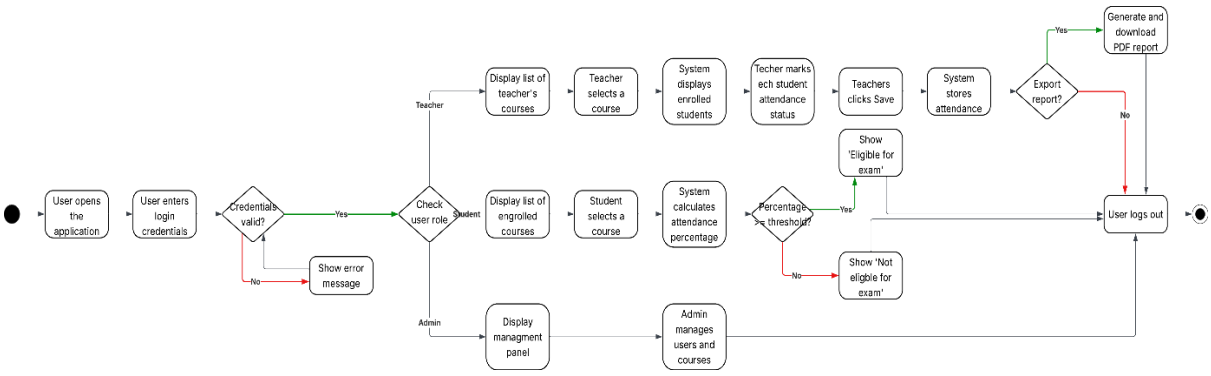

Figure 3 Activity Diagram - Represents the workflow of the system from user login to logout, including role-based routing, attendance marking, and report generation

5.1.4. State Diagram

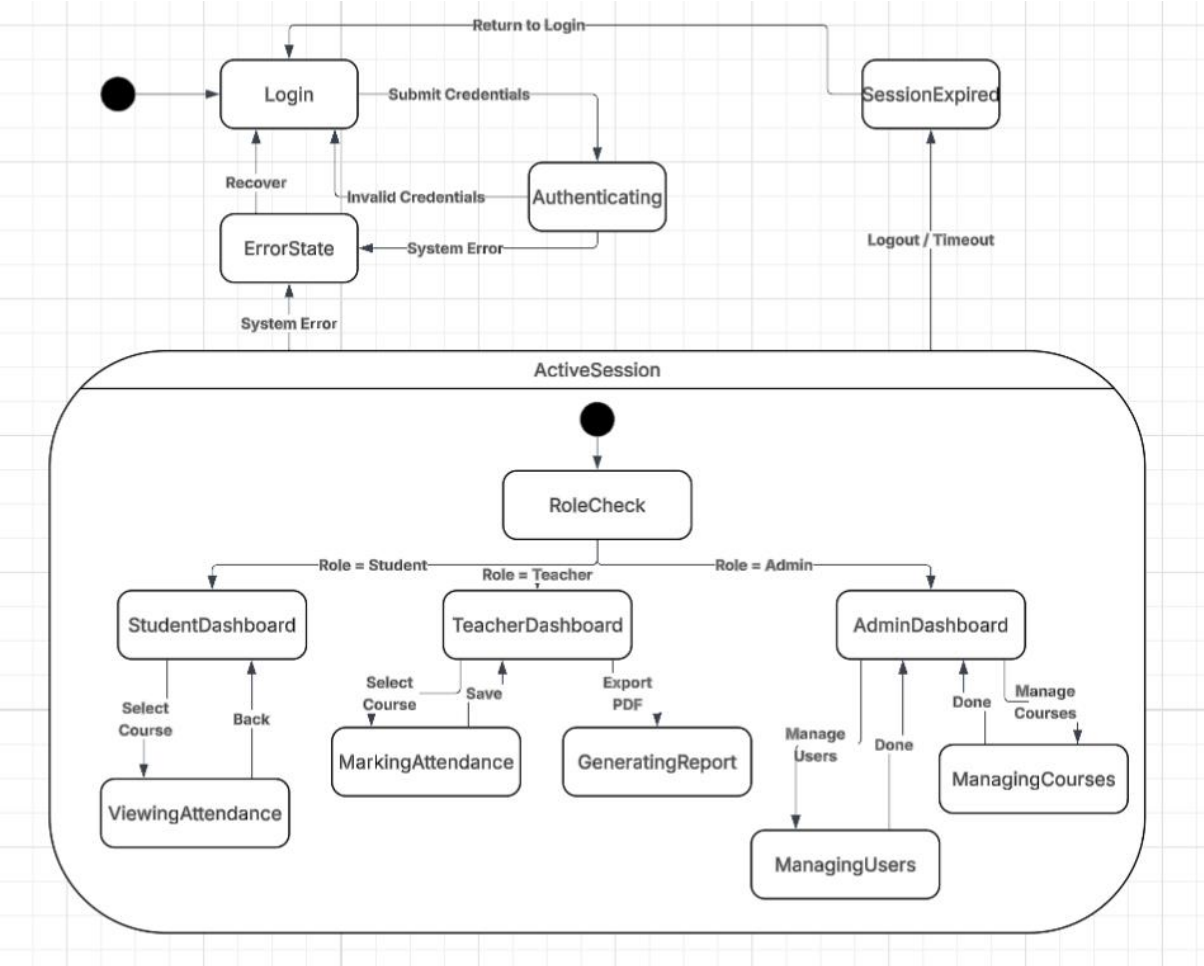

Figure 4 State Diagram - Depicts the different states of the system (Offline, Login, Authenticating, Active Session, Session Expired) and the transitions triggered by user actions and system events

5.1.5. Functional Requirements

FR-1.1: When entering valid login credentials, the system shall authenticate the user and allow access to the system

FR-1.2: When entering invalid login credentials, the system shall display an error message and deny access

FR-1.3: The system shall differentiate user roles (student or teacher) and display the appropriate content

FR-1.4: The system shall allow users to log out of the system securely

FR-1.5: When entering valid login credentials, the system shall authenticate the user and allow access to the system

FR-1.6: When selecting a course, the system shall display the list of students enrolled in that course

FR-1.7: The system shall store attendance records in the database after submission

FR-1.7: The system shall allow the teacher to mark students as present, absent or justified

6. Nonfunctional Requirements

6.1. Performance

NFR-1.1: The system shall respond to user requests withing 2 seconds under normal operating conditions

NFR-1.2: The system shall support at least 200 simultaneous users without performance drop

6.2. Security

NFR-2.1: The system shall require user authentication before accessing private data

NFR-2.2: The system shall restrict access to modifying attendance to authorized teachers only

7. GitHub repository

The GitHub repository - SE_Labs_Portfolio is meant to document the Lab work done during the semester for the Software Engineering class. It contains all the lab tasks completed during the classes as well as this document and a project. The repository is divided into three lab folders - LabTasks, Project and SRS_IEEE_Document. LabTasks foleder holds tasks produced during the respective classes with README.md files inside each sub directory that explains lab task. SRS_IEEE_Document directory holds together all versions of SRS documents

Repository URL: https://github.com/258564/SE_Labs_Portfolio